ADITYA RAJ

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SUMMARY

AI-focused software developer building and shipping production systems beyond coursework — including a live, deployed product.
Experienced across the full stack: frontend, serverless backend, LLM API integration, distributed systems, and applied ML pipelines.

EDUCATION

Vellore Institute of Technology *2025 – 2029*
B.Tech in Artificial Intelligence
Coursework: Data Structures · OOP · Machine Learning · Distributed Systems · Networking · Cyber Risk & Compliance

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript
Frontend: React, Vanilla JavaScript, TailwindCSS, Responsive UI
Backend & Systems: Flask, REST APIs, Serverless (Vercel), TCP/IP Networking, Concurrency
AI / ML: Scikit-learn, NumPy, Pandas, NLP, Semantic Search, LLM API Integration
Security: API key isolation, defensive JSON parsing, secure serverless deployment
Tools: Git, GitHub, VS Code, Vercel

FLAGSHIP PRODUCT — LIVE DEPLOYMENT

Baby Steps to the Mic — AI Public Speaking Coach *JavaScript · Web Speech API · Claude API · Vercel*
Independently designed, built, and deployed a production AI coaching platform — currently live.

- Built a Vanilla JS frontend with Web Speech API integration enabling real-time speech transcription and dynamic speaking topic generation per session
- Integrated Claude API via a Vercel serverless backend to deliver structured per-session feedback: scores, strengths, weaknesses, and coaching recommendations
- Implemented defensive JSON parsing to safely handle malformed or unpredictable LLM responses without crashing the session pipeline
- Isolated API keys server-side within serverless functions to prevent client-side exposure; managed environment configuration and production deployment end-to-end
- Made and executed all product decisions — architecture, frontend UX, backend API design, security posture, and deployment — from first commit to production

PROJECTS

Distributed Chat System *Java · TCP/IP · Concurrency*

- Built a multi-client TCP socket server in Java; assigned a dedicated thread per connection and synchronized shared state across concurrent sessions
- Implemented TCP/IP socket programming from scratch: server binds, accepts connections, and routes messages across active client sessions
- Applied thread synchronization primitives to prevent race conditions across shared data structures under simultaneous writes

DocuQuery — AI Document Retrieval System *Python · NLP · IR*

- Built an NLP-based document querying pipeline with structured extraction and relevance-ranked retrieval across multi-format inputs
- Built a preprocessing and indexing layer to normalize, tokenize, and vectorize documents into a queryable representation
- Tuned filtering and ranking logic to improve precision of returned results given noisy or ambiguous queries

Offroad-Semantic — ML Semantic Analysis *Python · Scikit-learn · NLP*

- Built a semantic classification pipeline: text ingestion, cleaning, TF-IDF feature extraction, vectorization, and Scikit-learn model inference
- Designed semantic mapping workflows to interpret context across unstructured inputs and assign structured categorical labels
- Tested multiple Scikit-learn classifiers and preprocessing combinations to identify the highest-accuracy configuration for the dataset

PassSecure — Flask Password Security Tool *Python · Flask · JavaScript*

- Built a Flask web app that evaluates password strength using entropy estimation, pattern matching (dictionary words, sequences, repeated chars), and character class checks
- Implemented secure input handling on both frontend and backend; designed responsive UI with real-time strength indicator and actionable feedback

Bank Management Simulation *Java · OOP*

- Designed a modular banking system with Account, Transaction, and Customer classes applying inheritance, encapsulation, and polymorphic dispatch
- Implemented deposit, withdrawal, transfer, and balance inquiry flows with structured transaction logging and edge-case validation

Library Management System *Java · SQL · CRUD*

- Built a Java application with full CRUD operations against a relational database: book inventory, member registration, issue/return tracking, and fine calculation
- Separated SQL data access from business logic into distinct layers, keeping the codebase modular and schema changes isolated

EXPERIENCE

Core Member — Bit By Bit Club, VIT *Dec 2025 – Present*

- Drive technical initiatives across collaborative builds; conduct peer code reviews focused on architecture quality and implementation decisions
- Participate in AI/ML-focused discussions and workshops bridging coursework concepts to applied engineering practice

Design & Content Creator — PranSpanda Club, VIT *Nov 2025 – Dec 2025*

- Led visual communication campaigns applying UI/UX principles to improve content structure, readability, and audience engagement across club platforms

Event Anchor & Coordinator *2023 – 2024*

- Anchored 5+ large-scale academic and cultural events; directed volunteer teams of 15+ and managed live audiences exceeding 500 attendees
- Developed structured public presentation and cross-functional coordination skills under real-time constraints

CERTIFICATIONS & ACHIEVEMENTS

- **Hackathon — Dawn of Code, VIT (MERN Stack Club):** shipped a full-stack prototype under time constraints
- **AWS Industry Workshop — Amazon Q in Code & Play:** hands-on cloud-native AI developer tooling
- **Machine Learning Foundations Certification**
- **Cyber Risk & Compliance Masterclass**
- **Deloitte Australia Data Analytics Job Simulation — Forage (Feb 2026):** built an interactive Tableau dashboard to analyze manufacturing downtime; applied Excel-based classification logic to categorize operational metrics and derive business insights
- **Deloitte Australia Cyber Security Job Simulation — Forage (Feb 2026):** analyzed 2,000+ lines of web activity logs to identify scripted API polling patterns and assess network exposure
- **National-Level Financial Literacy Quiz Participant**
- **International Academic Engagement — VLU Global Talk**